The longest chess game

The maximum length of a chess game under the FIDE Laws is 17,697 plies

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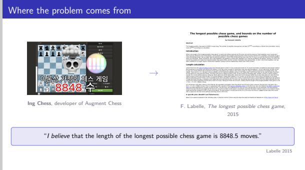
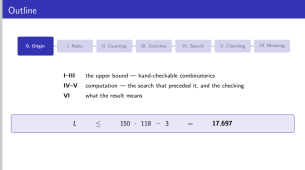
One sentence is the whole talk: the maximum length of a chess game is exactly 17,697 plies. The value has been known since 2015; what was missing is a proof that nothing is longer.



The starting point was a video by Ing Chess, the developer of Augment Chess, a chess-variant web game: “the theoretically longest chess game”.

Its description links to François Labelle’s page, and that page has essentially everything: the 118 delay plies, a three-switch schedule, a complete generated game.

And the conclusion reads “I believe”. The value was known; the proof was not. This talk is that proof.

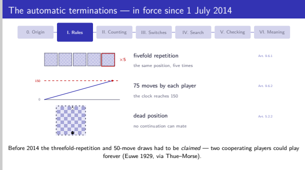


Seven parts, deliberately different in kind.

Parts I to III are pure mathematics and no computer appears in them. Parts IV and V are entirely about computation — the search that failed, and how the result was checked. Part VI is what it means.

Everything lands on that one line: a hundred and eighteen segments of a hundred and fifty, minus three. That minus three is the paper.

Statement numbers on these slides are the paper's own, so you can find any of them afterwards.



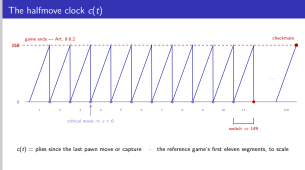
Before 2014 the question had no finite answer. Threefold repetition and the 50-move rule both had to be claimed, so two cooperating players could play forever — classically so, Euwe in 1929 by rediscovering the cube-free Thue–Morse sequence.

Since 1 July 2014 the Laws end a game automatically. No claim.

First: the same position five times. Second: 75 moves by each player with no pawn move and no capture — the moment the clock reaches 150. Third is the dead-position article that was already there, and bare kings are its clearest instance.

These rules have teeth. The seven-piece tablebases contain forced mates over five hundred moves long, and the second rule erases every one of them.

From that moment the longest game is a well-posed problem in extremal combinatorics.

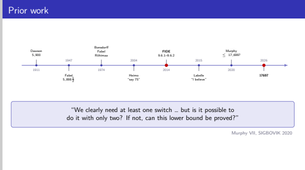


This picture is half the talk: the first eleven segments of the actual reference game, drawn to scale. The vertical axis is the halfmove clock. A pawn move or a capture resets it to zero; anything else pushes it up by one. The moment it touches 150 the game is over.

So a long game must move a pawn or capture something just before the clock would reach 150. Each of those resets is a critical move — Murphy's term; Labelle calls the same plies delay plies.

Look at the right-hand end. Only the last segment lets the clock actually reach 150, and the ply that gets it there is checkmate. That ending is legal only because of the precedence clause.

And segment eleven is 149. Why exactly one ply goes missing there is the whole of Part III.

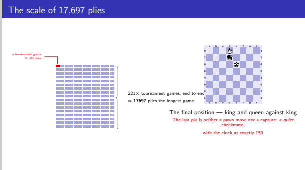


The question is older than the automatic rules. In the claim-based 50-move setting chess problemists worked it out over half a century: Dawson published 5,900 in 1911, Fabel settled it at 5,898 and a half in 1947.

In 2004 Heimo wrote something remarkable: with an odd bound — "say 75" — the answer would be White's 8,849th move. Ten years before the rule existed.

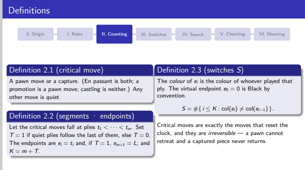
The rule arrived in 2014. In 2015 Labelle assembled everything and wrote "I believe". In 2020 Murphy published a hand-built game, computed a ceiling of 17,699 from the construction, and closed with the quoted question.

One switch is clearly needed, but can it be done with two? The answer today is no, and yes it can be proved.



For scale. A tournament game runs about forty moves, eighty plies. One red cell on the left is that, and the longest game is two hundred and twenty-one of them end to end.

On the right is where that game actually stops. King and queen against king. The last move is neither a pawn move nor a capture — a quiet checkmate played at the exact moment the clock reads 150. Without the precedence clause it would have been a draw.



Three definitions, numbered as in the paper.

2.1. A critical move is a pawn move or a capture. En passant is both, a promotion is a pawn move, castling is neither. These coincide exactly with the moves that reset the clock, and their irreversibility is decisive later.

2.2. The critical moves cut the game into segments. If quiet plies follow the last one, there is one closing segment more — that is T — and the number of segments is K.

2.3. The last ply of each segment is its endpoint, and its colour is whoever played it. S counts how often that colour changes, measured against a virtual Black endpoint at the start of the game. Without that convention the first segment could not be counted at all.

Lemma 2.4 — the segment decomposition

Lemma 2.4 (segment decomposition)

$$L \;=\; 150K - S - \sum_{i=1}^{K} \delta_i, \qquad \delta_i \geq 0$$

δ_i is segment i's shortfall from its target length (150 if its endpoints share a colour, 149 otherwise). In particular $L \leq 150K - S$.

Proof

(1) All plies of segment i except possibly the last are quiet. If its length were 151 or more, the first 150 would all be quiet and the game would already have ended there (Art. 9.6.2) — contradiction. So the length is ≤ 150.

(2) Plies alternate colour, so a segment's length is even exactly when its two endpoints share a colour. If they differ the length is odd, and being ≤ 150 it is ≤ 149.

(3) Summing the segment lengths telescopes to $e_K - e_0 = L$. □

Lemma 2.4 is the whole talk's ledger.

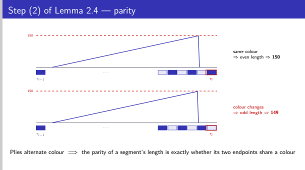
The proof has three steps.

Step one: a segment cannot exceed 150 plies. If it did, its first 150 would all be quiet and 9.6.2 would already have ended the game there.

Step two is where S comes from, and the next slide shows it as a picture.

Step three is a telescoping sum.

The conclusion is $L \leq 150K - S$. Now bound K and S separately.



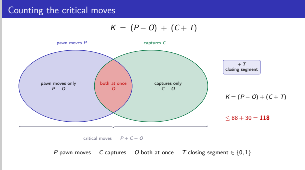
This is pure parity.

The picture uses the same idiom as the clock graph: over one segment the clock climbs and the final critical move drops it to zero. The strip below the axis is the colour of each ply.

Top row: if the two endpoints share a colour, the number of plies between them is even. 150 is even, so the segment can be filled.

Bottom row: if they differ, the length must be odd, and the largest odd number at most 150 is 149. So each colour change costs exactly one ply.

That is the entire reason S appears in the decomposition.

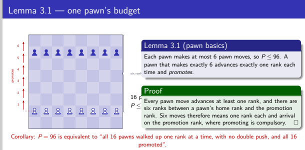


Count the critical moves by inclusion–exclusion.

A critical move is a pawn move or a capture, so the total is pawn moves plus captures minus the ones that are both. Those are the diagonal pawn captures, the overlaps O. Add one if there is a closing segment.

The grouping matters: writing it as $P - O$ plus $C + T$ lets the two halves be bounded completely independently.

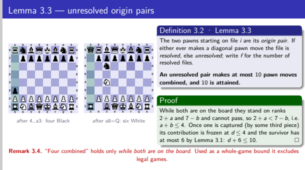
And it has to hold for *every* game. Reading “eight overlaps, so 96 minus 8” off the known construction is circular — whether that construction is optimal is exactly what we are proving. The next four slides do the work properly.



The easy one first. Six ranks between a pawn's home rank and promotion, and every pawn move advances at least one. So six per pawn, sixteen pawns, $P \leq 96$.

The corollary matters more. If a pawn made exactly six moves then each of them advanced exactly one rank — no double pushes — and it finished on the promotion rank, where promoting is compulsory.

So $P = 96$ says: all sixteen pawns walked up one square at a time and all sixteen promoted. It also says no pawn is left on the board. We use it repeatedly.

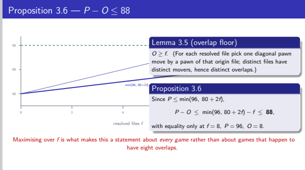


The two pawns that start facing each other on a file are that file's origin pair. If neither ever moves diagonally the file is unresolved, and f counts the resolved files.

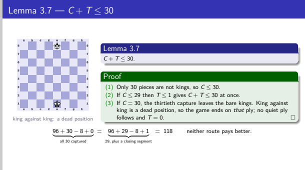
While both are on the board they cannot pass. Capturing each other would need a diagonal move, excluded by assumption, and neither can overtake — you cannot push onto an occupied square nor jump one with a double push. So four combined moves at most.

But if one of them is captured by some *other* piece, the survivor is free. On the left Black's a-pawn walks four squares and is taken by a knight; on the right White's a-pawn walks the whole file and promotes. Ten, not four.

Remark 3.4 is in the paper for a reason: my own model got this wrong at one stage. Using four as a whole-game bound excludes legal games.



Two ceilings.
One: ten per unresolved file and twelve per resolved one, so $10(8 - f) + 12f = 80 + 2f$. The solid line.
Two: six per pawn, 96 outright. The dashed line.
But every resolved file contains at least one diagonal pawn move, which is a capture and hence an overlap, and different files contribute different moves. So $O \geq f$: resolving a file costs an overlap.
Subtract and you get the thick line, peaking at 88 when f is eight and strictly smaller everywhere else.
Maximising over f is the decisive step. Without it this would be a claim about one game rather than all of them.

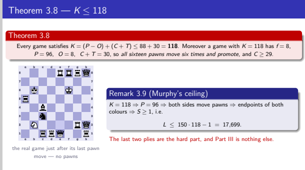


At most thirty captures, since kings are not capturable.

But taking all thirty leaves the two kings alone, and king against king is a dead position — no sequence of legal moves mates. So the game ends the instant it arises, nothing quiet can follow, and $T = 0$.

Hence $C + T \leq 30$. As the display shows, taking all thirty and losing the closing segment, or taking twenty-nine and gaining one, both give 118.

Worth noting: this is the only place dead-position reasoning enters the bound at all, and king against king is the one case where deadness is beyond dispute.



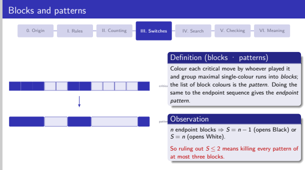
Putting it together gives 118.

The equality case is the useful part. To hit $88 + 30$ both halves must be tight, and Proposition 3.6's equality case pins f at eight, P at 96, O at eight.

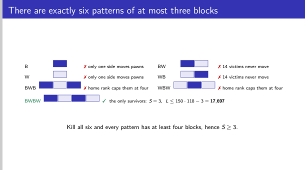
As we saw, $P = 96$ means all sixteen pawns promoted. On the left is the real game just after its last pawn move: no pawns, and promoted rooks and queens everywhere. And $T \leq 1$ gives $C \geq 29$.

Remark 3.9 comes free. $P = 96$ means both sides move pawns, a pawn move is a critical move by its owner, so endpoints of both colours occur. The virtual endpoint is Black and some endpoint is White, so the colour changes at least once — 17,699, Murphy's ceiling, now a theorem about all games.

Remember two things: $P = 96$ and $C \geq 29$. Every argument in Part III runs on them.



One more piece of vocabulary.
Colour each critical move by whoever played it — a pawn move by the pawn's owner, a capture by the capturer.
That gives a sequence of colours; group maximal runs of one colour into blocks and list the block colours, and you have the pattern. Top left is the sequence, below it the pattern.
Do the same for endpoints and you get the endpoint pattern. With n blocks, S is $n-1$ if it opens Black and n if it opens White, because then it also disagrees with the virtual Black endpoint.
So ruling out $S \leq 2$ is exactly ruling out every pattern of at most three blocks.

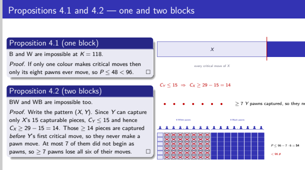


A nonempty alternating pattern of at most three blocks is one of exactly six: B, W, BW, WB, BWB, WBW. That is the complete list.

Show that none is possible at $K = 118$ and every pattern has at least four blocks, hence $S \geq 3$.

The first four die by counting alone. The two three-block patterns are harder and need a lemma.

To spoil the ending: the survivor is BWBW, and that is exactly the pattern of the real game.



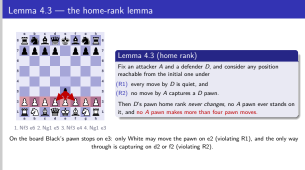
Proposition 4.1 is a single line. If only one colour ever makes a critical move then only its eight pawns ever move: $P \leq 48$, and we need 96.

Proposition 4.2 is more interesting. Write the pattern (X, Y): every critical move of X precedes every critical move of Y.

Y can capture only X's pieces, fifteen of them, so $C_Y \leq 15$. Since $C \geq 29$, that leaves $C_X \geq 14$. The side moving first makes at least fourteen captures, all before Y has made a single critical move.

So those pieces were off the board before Y's first critical move, which means they never made a pawn move at all. At most seven of them did not begin as pawns — queen, two rooks, two bishops, two knights — so at least seven pawns lose all six of their moves.

That is the grid: forty-two cells gone, $P \leq 54$, nowhere near 96.



This is the heart of the paper.

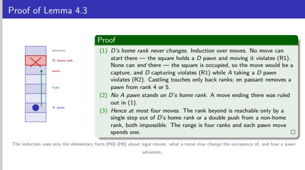
The setting: an attacker A and a defender D, where D plays only quiet moves and A never captures a D pawn.

On the board, Black's pawn has walked four squares to e3 and stopped. It has nowhere to go.

There is a White pawn on e2. Only White may move it, and moving a pawn is not a quiet move, so R1 forbids it. Black cannot push through because the square is occupied. Going round means capturing on d2 or f2, which is exactly what R2 forbids.

So the home rank becomes a wall. Not one square of it ever changes.

The rest of the proof, and why exactly four, is the next slide.



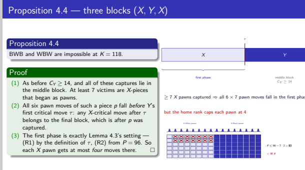
Once the wall is fixed, the rest is arithmetic.

Step one is the wall. No move can start on those squares and none can end there. Castling and en passant are checked separately.

Step two: an *A* pawn cannot stand on *D*'s home rank, because a move ending there was just excluded.

Step three: it cannot get past either. The rank beyond needs a single step out of *D*'s home rank — which it can never occupy — or a double push from somewhere that is not its own home rank, which the rules forbid.

So the range is the four ranks between its home rank and *D*'s, and each pawn move spends at least one: four moves at most. And four is attained.



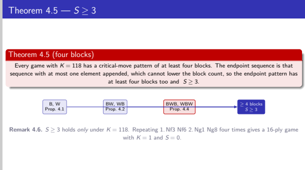
Now the three-block case.

Step one is as before: $C_Y \geq 14$, and all of Y's critical moves lie in the middle block, so all of those captures do too. At least seven victims are X pieces that began as pawns.

Step two: take one of them, p. It must make six pawn moves, all before it is captured. But any critical move by X after Y's first one — ply τ — belongs to the final X block, which comes after p was taken. A piece off the board cannot move. So all six fall in the first phase.

Step three: the first phase is exactly the lemma's setting. R1 is immediate from the definition of τ; R2 comes from $P = 96$, because a captured Y pawn would make no pawn move at all.

So each X pawn gets at most four moves where six are needed. Seven pawns short by two gives $P \leq 82$.



All six are dead, so the pattern has at least four blocks.

The endpoint sequence is the critical-move sequence with at most one element appended, and appending cannot reduce the block count. So the endpoint pattern also has at least four blocks, and $S \geq 3$.

Remark 4.6 deserves emphasis. $S \geq 3$ is conditional on $K = 118$ and nothing else. In general it is plainly false: repeat Nf3 Nf6 Ng1 Ng8 four times and the game ends after sixteen plies by fivefold repetition of the initial position, with no pawn move and no capture, so $K = 1$ and $S = 0$.

That costs nothing, because a game with $K \leq 117$ has already forfeited a whole segment and no switch count makes that back.

Theorem 5.1 — the upper bound

Theorem 5.1 (upper bound)

No legal chess game lasts more than 17,697 plies.

Proof

$K \leq 117$ Lemma 2.4 with $S \geq 0$: $L \leq 150 \cdot 117 = 17{,}550 < 17{,}697$

$K = 118$ the largest Theorem 3.8 permits; Theorem 4.5 gives $S \geq 3$, so $L \leq 150 \cdot 118 - 3 = \mathbf{17{,}697}$

□

Corollary 5.3 (exact maximum)

The bound is attained (Labelle's generated game, Murphy's hand-built one). **The maximum length of a chess game is exactly 17,697 plies — 8,848 moves and one final White ply.**

The proof is two lines.
If $K \leq 117$ a whole segment is missing, so $L \leq 150 \times 117 = 17{,}550$, less than 17,697.
If $K = 118$, the theorem we just proved gives $S \geq 3$, so $L \leq 150 \times 118 - 3 = 17{,}697$.
Keeping the cases separate is the point: $S \geq 3$ is not available below 118 and is not needed there either — one segment outweighs any switch count.
And the lower bound already exists: two published games are exactly this long. So the maximum is exactly 17,697 — an equality, not an inequality.

Corollary 5.2 — the structure of a maximum-length game

Corollary 5.2

Every game of length 17,697 satisfies $K = 118$, $S = 3$, every $\delta_i = 0$, $f = 8$, $P = 96$, $O = 8$, $C + T = 30$, and both its critical-move and endpoint patterns are BWBW.

all sixteen pawns move six times, one rank each, and promote · every segment is full (150, or 149 across a change)

BWBW Black makes the first critical move, and the final endpoint is White's.

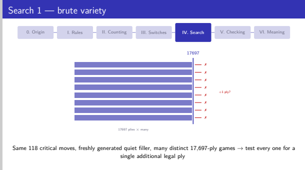
The equality analysis costs nothing extra.

If the length is 17,697 then every inequality above was an equality. So $K = 118$, S is exactly three, and every δ is zero — not one segment wasted.

All the equality conditions follow: sixteen pawns promoted, exactly eight overlaps.

The pattern is pinned too. There are exactly four blocks, and it cannot open with White — that would disagree with the virtual Black endpoint as well and force $S \geq 4$. So it is BWBW.

Which means we know what a maximum-length game looks like. Black makes the first critical move; White plays the last. There is no choice about it.



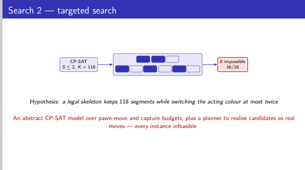
Chronologically this came first. I was trying to break 17,697.

The first attempt was the blunt one: start from the known game, keep its 118 critical moves, regenerate all the quiet filler between them, and mass-produce distinct 17,697-ply games. One of them differs from the original in 17,032 of its plies.

Then ask every one of them the same question: can you add one more legal ply?

Never. Every game hit exactly the same wall.

That is not a proof, but it was the signal that the wall was not an accident.

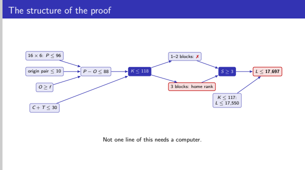


The second attempt had a target: there is a legal skeleton with 118 segments that switches the acting colour at most twice.

I built an abstract CP-SAT model over the pawn-move and capture budgets, and a concrete planner to turn any candidate into an actual sequence of moves.

Every instance came back infeasible. The planner never had anything to do.

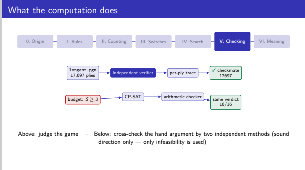
That is where I turned around. Failing to find something repeatedly can mean it is not there. I stopped treating 17,697 as a target to beat and started treating it as a theorem to prove.



And this is the shape the proof took. On the left the elementary lemmas about pawns and captures, in the middle $K \leq 118$, on the right the pattern argument and $S \geq 3$.

The box underneath is the $K \leq 117$ case, which simply runs out of segments at 17,550.

The point: none of this needs a computer. All of it is checkable by hand. The computer appears in the next part, and what it does there is cross-check, not replace.



The computer does two things.

Above the line: an independent verifier replays Murphy's game from the initial position, ply by ply. Legality, whether anything terminates early, a classification of the position after every ply — and it passes a sequence only if the game ends exactly on its last move. That verifier imports nothing else from the repository, so the judge shares no code with any producer of games.

Below the line: every finite case analysis is exhausted mechanically, and the pattern question — does any endpoint pattern of at most three blocks admit $K = 118$? — is decided twice more, by a CP-SAT model and by a solver-free arithmetic checker that declares its own constants.

Both are used in the sound direction only. Infeasible means no such game exists; feasible does not mean one does. No claim in the paper rests on a feasibility verdict.

Two rules the verifier implements

Mate outranks the clock

The final ply is the 150th quiet ply of its segment *and* a checkmate (the precedence clause of Art. 9.6.2).

A verifier that consults the clock first accepts the same 17,697 plies but scores the game a draw.

Never compare whole FENs

For repetition the clock and move number play no part in identity, and en passant counts only when *legal* (Art. 9.2).

Naive keys fail both ways — a FEN over-distinguishes; placement plus side to move ignores castling rights and over-merges.

Under that identity no position of this game occurs even *three* times — fivefold repetition is never approached.

These are rules of chess, not engineering choices, and the reference game exercises both.

First, the order of the termination tests. The final ply is simultaneously the 150th quiet ply of its segment and a checkmate; the precedence clause keeps the mate. A verifier that looks at the clock first accepts the same 17,697 plies but records a draw — right length, wrong result.

Second, position identity for repetition. Do not compare whole FEN strings: the halfmove clock and move number play no part in identity, and the en passant target is written after every double push whether or not a capture is legal, so fivefold would never fire. Comparing only placement and side to move goes wrong the other way, ignoring retained castling rights.

As an aside, Tom 7's reference implementation still has MOVE_RULE set to 50 directly beneath a comment saying 75 is correct. A reference implementation is not an oracle — which is why ours is independent.

Verdict and reproduction

the game	legal, 17,697 plies, ends in checkmate
early termination	none — no position occurs even three times, no stalemate
profile	$K = 118,\ P = 96,\ C = 29,\ O = 8,\ T = 1,\ S = 3$ — exactly Corollary 5.2
decomposition	$115 \times 150 + 3 \times 149 = 17{,}697,\ \sum_i \delta_i = 0$
pattern	endpoint pattern BWBW
cross-check	all six patterns of ≤ 3 blocks infeasible, both methods agree

A second game is verified the same way — same 118 critical moves, fresh filler, 17,032 plies differ.

Everything re-runs from a byte-for-byte reproducible certificate: `make verify-full`

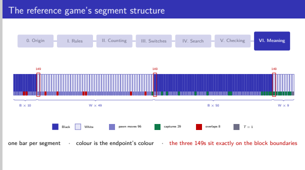
The verification in one table.

The game is legal, 17,697 plies, ends in checkmate. Nothing terminates early — the clock stays under 150 after every ply but the last, and no position ever occurs even three times.

And the profile is exactly what Corollary 5.2 forces. The decomposition closes exactly: 115 segments of 150, three of 149, nothing wasted.

Here is the thing worth saying out loud: improving on such a game is not hard, it is impossible. Every segment is full, and the three plies it loses are exactly the three the theorem proves unavoidable.

So that the result does not rest on a single artefact, a different game of the same length was rebuilt and verified the same way.



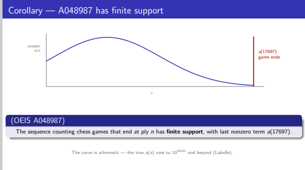
This is the game. One bar per segment, all 118, taken from the actual PGN rather than drawn as a schematic.

A bar's colour is the colour of that segment's endpoint, so the four visible runs are the BWBW pattern: ten Black, forty-nine White, fifty Black, nine White.

The strip along the bottom says what ended each segment — a pawn move, a capture, or both. Ninety-six pawn moves, twenty-nine captures, eight overlaps, one closing segment. The whole count, visible.

And the three marked in red are the 149s. Look at where they are: exactly on the block boundaries. Segments eleven, sixty and a hundred and ten. One ply lost wherever the colour changes.

What we proved today is that those three cannot be two.

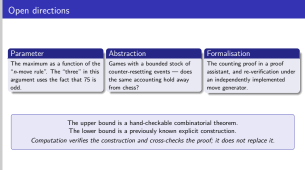


Another way to state the result.

OEIS A048987 counts chess games that end at ply n. What we proved is that this sequence has finite support, and that its last nonzero term is exactly $a(17697)$.

Read the curve for shape only. The real values are enormous — Labelle's Monte Carlo estimate puts the total above 10^{29241}. Treat the vertical axis as "nonzero".

That chess is finite has been known since 2014. What today adds is where it stops.



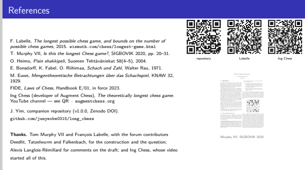
Three directions and then I will stop.

One. Replace 75 by n and ask for the answer as a function of n. The "three" here uses that 75 is odd; the even case is separate.

Two. Abstraction: games with a finite stock of counter-resetting events — does the same accounting go through away from chess?

Three. Formalisation. This is a counting proof, a good shape for a proof assistant. And we currently trust python-chess's move generation; re-verifying under an independent implementation would shrink that trust.

One last sentence. The upper bound is a hand-checkable theorem, the lower bound is a construction that already existed, and computation verifies the construction and cross-checks the proof without replacing it. Thank you.



References and thanks.

This work exists because Tom Murphy VII built the game and asked the question. The numbers behind the answer — the 118 and the three-switch structure — were found years earlier by François Labelle and the forum contributors his page credits. What I added is the proof that it is optimal.

The three codes are the repository, Labelle's page, and the Ing Chess channel where this problem started. The repository has the paper, the verifier and a reproducible certificate; one make verify-full re-runs all of it in about twenty seconds.

Questions.